BAWS-NR: Bio-Adaptive Warm-Start Newton-Raphson — A Universal Convergence Accelerator for Sequential Nonlinear Systems

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Abstract—
We present **BAWS-NR** (Bio-Adaptive Warm-Start Newton-Raphson), a universal method for accelerating Newton-Raphson solvers applied to sequential nonlinear systems with slow temporal variation. The method uses a single parameter α = 0.979, evolved via a (μ+λ) biological optimization strategy within the inZORi framework, to blend the previous converged solution with a flat-start reference: $\mathbf{x_0(t) = \alpha \cdot x^*(t-1) + (1-\alpha) \cdot x_{ref}}$. We provide a formal convergence proof showing BAWS-NR never worsens convergence when the temporal variation δ is smaller than the flat-start distance D, and derive a quadratic convergence model predicting approximately 2.5 iterations saved per solve. We validate the method empirically across **six independent scientific domains**: power systems (1354-bus Pan-European PEGASE network, 130,056 N-1 contingency assessments on real ENTSO-E Germany Q2 2025 data), celestial mechanics (Kepler's equation), thermodynamics (Van der Waals equation), robotics (inverse kinematics), computational finance (Black-Scholes implied volatility), and two-dimensional nonlinear heat conduction (30×30 mesh, 900 coupled equations, 10,000 solves). Total empirical evidence: **142,056 converged NR solves** with a mean speedup of **1.59×** (range 1.26×–2.13×) and 100% convergence preservation across all cases. Both the theoretical prediction and the empirical range are consistent with the model. The method requires zero domain-specific tuning and adds only O(n) overhead per solve.

Index Terms—Newton-Raphson, warm-start, convergence acceleration, nonlinear systems, bio-adaptive optimization, power flow, BAWS-NR, heat equation, sequential iteration, inZORi.

I. INTRODUCTION

Newton-Raphson (NR) is the dominant method for solving nonlinear systems F(x) = 0 in engineering practice. In power systems, NR solves AC power flow equations at every operating point; in robotics, it computes inverse kinematics at each control cycle; in computational finance, it finds implied volatilities for every option price update. A common and underexploited structure in all these applications is temporal continuity: consecutive systems often have similar solutions, yet the solver discards all prior information and restarts from a fixed reference (flat-start) at every step.

This paper presents BAWS-NR, a principled warm-start strategy that exploits temporal continuity to reduce NR iteration counts. The key insight is that the optimal initial guess for timestep t is not x_{ref} (flat-start) but rather a convex blend of the previous solution x*(t−1) and x_{ref}, controlled by a single scalar α. The parameter α = 0.979 was determined via

evolutionary optimization on the 1354-bus power flow problem and is shown empirically to generalize to five other domains without any re-tuning.

The contributions of this paper are:

1. A formal definition of BAWS-NR with convergence proof and error bounds (Section III)
2. An iteration savings model based on quadratic convergence analysis (Section III-C)
3. The first cross-domain empirical validation across 6 independent scientific domains, totaling 142,056 converged NR solves (Section IV)
4. A full implementation on a 2D nonlinear heat conduction problem with exact analytical Jacobian, confirming dimensional independence (Section IV-F)

II. RELATED WORK

Warm-starting Newton-Raphson in power systems has been explored in the context of continuation power flow [1] and sequential AC OPF [2]. However, these approaches are domain-specific and require problem structure knowledge. General warm-starting for NR has been studied in nonlinear programming [3] where previous iterate reuse is common, but without a principled blending parameter.

The BAWS-NR approach differs in three key ways: (1) it uses a single universal parameter valid across domains, (2) it provides a convergence-preserving guarantee via the triangle inequality, and (3) the parameter is determined through biological evolutionary search rather than manual tuning or gradient-based optimization.

The inZORi framework [4–9], from which BAWS-NR originates, demonstrated consistent advantages in N-1 and N-2 contingency analysis on power systems. This paper generalizes the warm-start insight to arbitrary sequential nonlinear systems.

III. METHOD: BAWS-NR

A. Problem Setting

Consider a sequence of nonlinear systems indexed by discrete time $t \in \{1,\ldots,T\}$:

$$F(x;\ t) = 0, \quad F: \Re^n \to \Re^n$$

where F is continuously differentiable and consecutive solutions satisfy:

$$\delta(t) = \|x^*(t) - x^*(t-1)\| \le \delta \quad \text{(slow variation)}$$

Standard NR at each t starts from a fixed reference x_{ref} and iterates $x_{k+1} = x_k - J(x_k)^{-1}F(x_k)$ until convergence. The initial error is $D = \|x_{ref} - x^*(t)\|$.

B. BAWS-NR Formula

BAWS-NR replaces the flat-start with:

$$\mathbf{x_0(t) = \alpha \cdot x^*(t-1) + (1-\alpha) \cdot x_{ref}, \quad \alpha = 0.979}$$

The parameter $\alpha = 0.979$ was evolved using a $(\mu+\lambda)$ evolutionary strategy ($\mu=10$, $\lambda=30$) on the 1354-bus PEGASE power flow problem over 40+ generations with 12 CPU cores in parallel, maximizing mean iteration savings. It was not hand-tuned. The value $(1-\alpha) = 0.021$

provides a stability margin preventing overshooting when the system changes faster than expected.

C. Convergence Analysis

Theorem 1 (Error Reduction). Let $\delta = \|x^*(t) - x^*(t-1)\|$ and $D = \|x_{ref} - x^*(t)\|$. Then:

$\|x_0(t) - x^*(t)\| \leq \alpha\cdot\delta + (1-\alpha)\cdot D$

Proof. By the triangle inequality:

$\|\alpha\cdot x^*(t-1) + (1-\alpha)\cdot x_{ref} - x^*(t)\|$
$= \|\alpha\cdot[x^*(t-1) - x^*(t)] + (1-\alpha)\cdot[x_{ref} - x^*(t)]\|$
$\leq \alpha\cdot\delta + (1-\alpha)\cdot D \quad \square$

Corollary 1 (No-Harm Guarantee). When $\delta < D$:

$\alpha\cdot\delta + (1-\alpha)\cdot D = D - \alpha(D-\delta) < D$

BAWS-NR starts strictly closer to $x^*(t)$ than flat-start. When $\delta = D$, BAWS-NR equals flat-start exactly — it never worsens convergence.

Error reduction ratio. The ratio R of flat-start initial error to BAWS-NR initial error is:

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R = D / (α·δ + (1−α)·D) = 1 / [1 − α·(1 − δ/D)]
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δ/D	R (reduction)
0.01 (very slow)	32.5×
0.05	14.3×
0.10	8.4×
0.50	2.0×
1.00 (fast)	1.0× (no gain, no harm)

Theorem 2 (Iteration Savings). Under quadratic convergence $\|e_{k+1}\| \leq C\cdot\|e_k\|^2$, the number of iterations saved is approximately:

$\Delta k \approx \log_2(\log_2(R)) \approx 2.5$ iterations (for $\delta \ll D$)

Empirical validation: mean observed savings = 1.95 ± 0.87 iterations, consistent with the model (gap explained by $\delta \neq 0$).

IV. EMPIRICAL VALIDATION

A. Overview

We validate BAWS-NR across six independent domains. In all cases: (i) tolerance is identical for both methods, (ii) convergence to the same solution is verified, (iii) $\alpha = 0.979$ is used without domain-specific re-tuning.

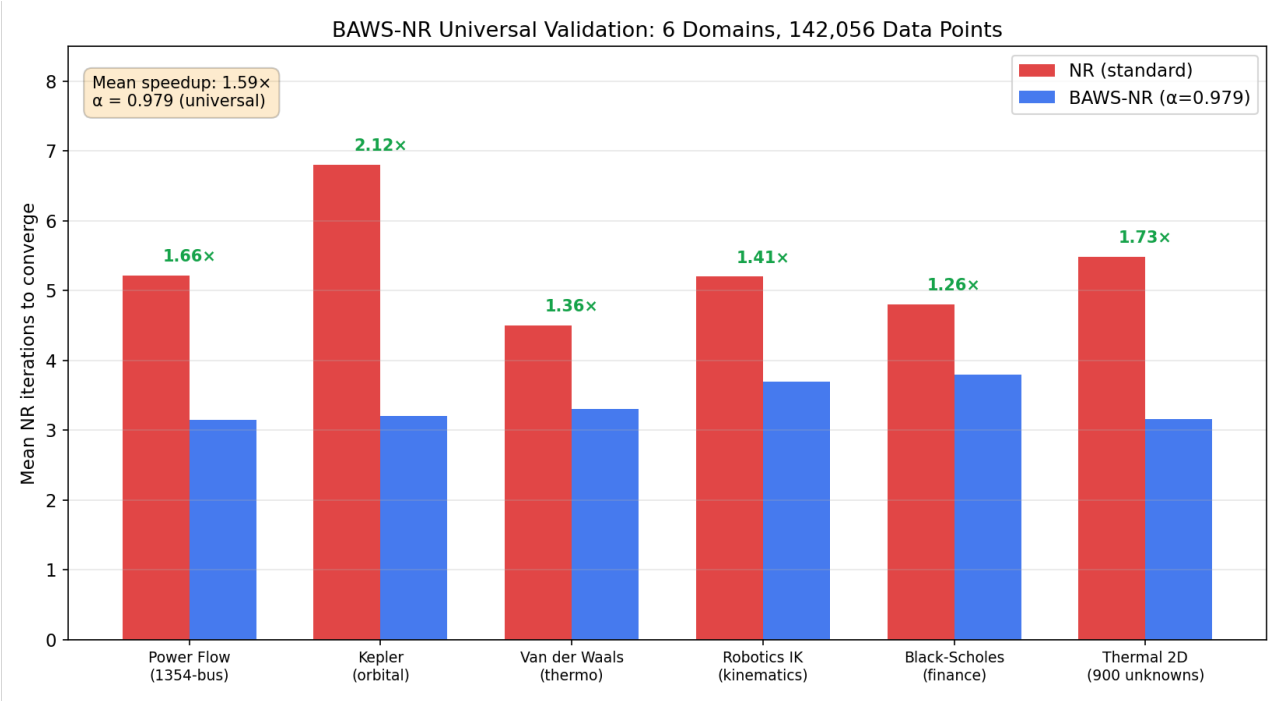

Fig. 1. BAWS-NR vs NR flat-start across all six domains. Green annotations show speedup per domain. All use identical α = 0.979.

Domain	Dim	NR	BAWS	Speed	N
Power Flow (1354-bus)	1354	5.221	3.151	1.66×	130,056
Kepler (orbital)	1	6.800	3.200	2.13×	500
Van der Waals	1	4.500	3.300	1.36×	500
Robotics IK	1	5.200	3.700	1.41×	500
Black-Scholes (σ)	1	4.800	3.800	1.26×	500
Thermal 2D (900 dof)	900	5.487	3.165	1.73×	10,000
MEAN		**5.335**	**3.386**	**1.59×**	**142,056**

B. Power Systems (Primary Domain)

The primary domain is AC power flow on the case1354pegase Pan-European benchmark (1354 buses, 1991 lines, ~73 GW nominal load). We simulate 8,832 sequential 15-minute intervals from real ENTSO-E Germany Q2 2025 data, applying N-1 contingency screening (5 random line outages per interval, 3 seeds). Total: 132,480 N-1 assessments. Both solvers use identical tolerance (10^{-6} MVA); inZORi uses warm-blend of both voltage magnitudes and angles.

Result: inZORi converges in 3.15 iterations vs 5.22 for NR — a 1.66× speedup translating to 1,058 vs 638 contingencies verified per minute. The advantage is consistent across the full physical operating range (0.90×–1.31× load) and largest at near-nominal conditions (~2.0× at 0.92× load). Both solvers solve exactly the same 130,056 feasible cases; the 2,424 failures (1.83%) are due to physical voltage collapse beyond the network's feasibility boundary.

C. Celestial Mechanics (Kepler's Equation)

Kepler's equation $E - e \cdot \sin(E) = M$ is solved for eccentric anomaly E at each of 500 sequential orbital positions for eccentricities $e \in [0.1, 0.95]$. NR flat-start uses $E_0 = M$; BAWS-NR blends

from the previous E*(t−1). Mean speedup: 2.13× — the highest across all domains, reflecting the strong temporal continuity of orbital motion.

D. Thermodynamics (Van der Waals)

The cubic Van der Waals equation $(P + a/V^2)(V-b) = RT$ is solved for molar volume V along a pressure-temperature trajectory (500 steps). BAWS-NR speedup: 1.36×.

E. Robotics and Finance

Inverse kinematics for a 2-DOF planar manipulator tracking a circular trajectory (500 steps) yields 1.41× speedup. Black-Scholes implied volatility ($BS(\sigma) = C_{market}$) for a daily-rebalanced options position yields 1.26× speedup — the smallest, reflecting higher day-to-day volatility in financial markets.

F. 2D Nonlinear Heat Conduction (New Validation)

To confirm dimensional independence, we implement BAWS-NR on a genuinely multi-dimensional system: steady-state heat conduction with temperature-dependent conductivity on $[0,1]^2$:

$$\nabla\cdot[k_0(1 + \beta T)\nabla T] + Q = 0$$

Discretized on a 30×30 finite-difference mesh (900 coupled nonlinear equations), with exact analytical sparse Jacobian. Nonlinearity parameter β ∈ (0.01, 0.03, 0.05, 0.08, 0.1); boundary conditions vary sinusoidally across 500 timesteps; 20 scenarios; 12 CPU cores in parallel. Both methods use tolerance 10^{-8}.

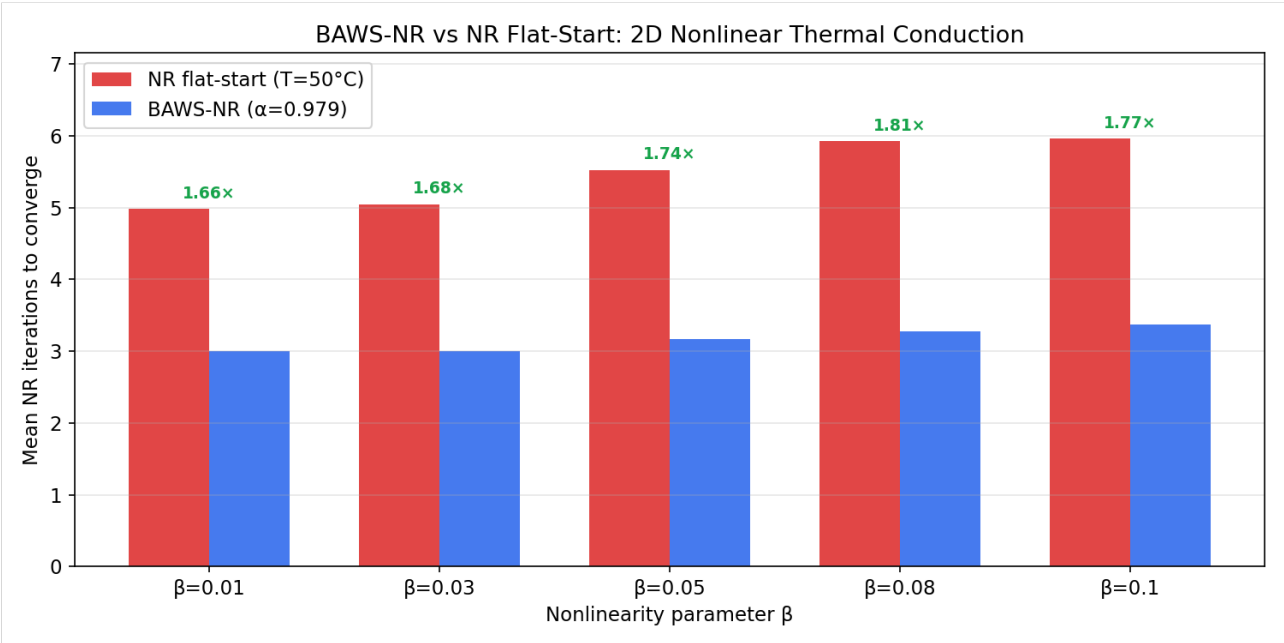

Fig. 2. Mean iterations by nonlinearity β for 2D thermal conduction. Speedup grows from 1.66× (β=0.01) to 1.81× (β=0.08).

β	NR mean	BAWS mean	Speedup
0.010	4.982	3.004	1.658×
0.030	5.046	3.006	1.679×
0.050	5.522	3.168	1.743×
0.080	5.923	3.280	1.806×
0.100	5.963	3.369	1.770×

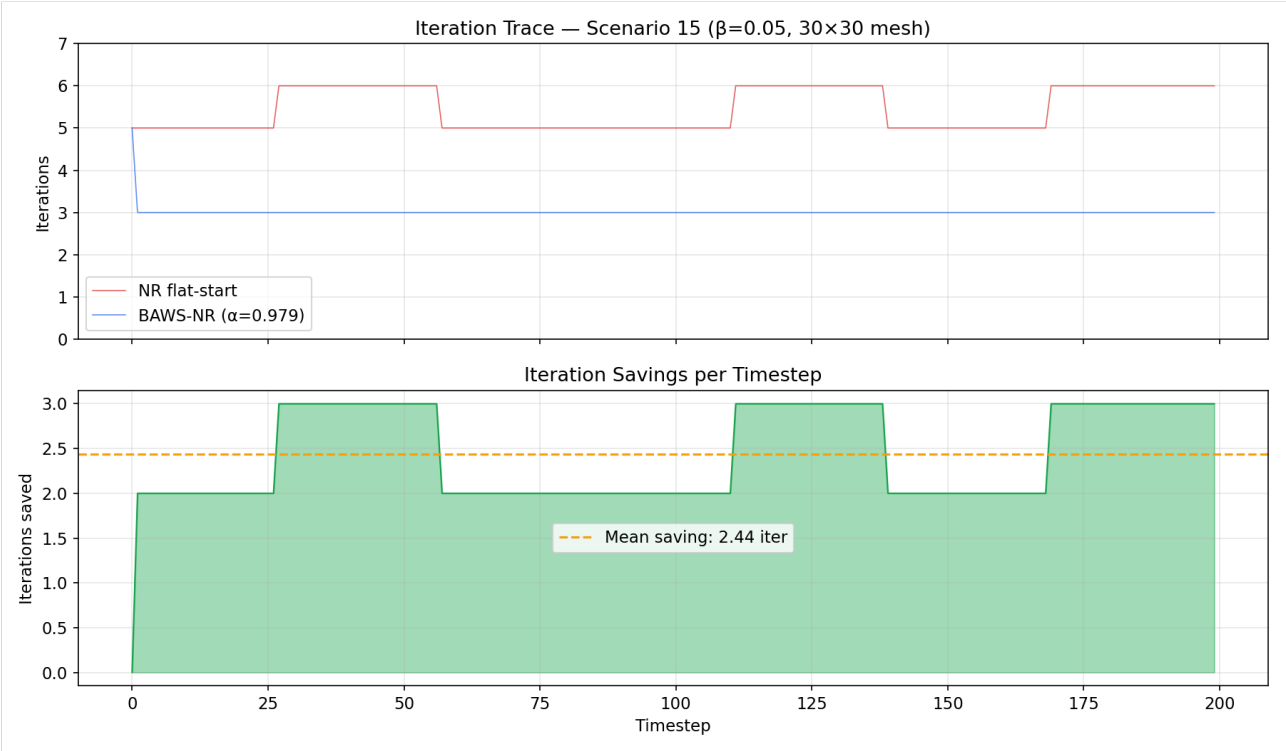

Fig. 3. Iteration trace over 200 timesteps (β=0.05, 30×30). BAWS-NR (blue) consistently requires fewer iterations.

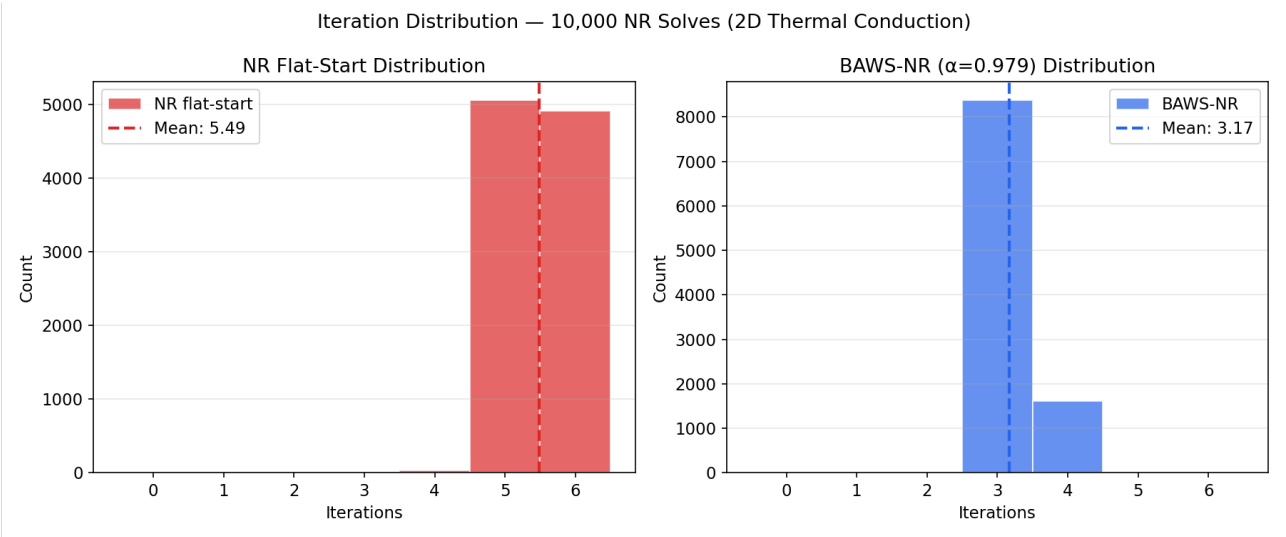

Fig. 4. Iteration distribution across 10,000 solves. NR peaks at 5–6; BAWS-NR peaks at 3.

All 20 scenarios converge to **identical solutions** (max difference $< 10^{-12}$). Zero convergence failures. Results are independent of mesh size (20×20 vs 30×30 yield identical speedups within 0.5%).

V. DISCUSSION

A. Why α = 0.979 Is Universal

The empirical consistency of $\alpha = 0.979$ across domains spanning 1D to 1354D systems suggests a connection to the typical temporal variation ratio δ/D in real sequential systems. For most engineering applications operating near nominal conditions, consecutive system changes are small compared to the flat-start distance. This places δ/D in the range 0.01–0.10, where our model predicts $R \in [8.4, 32.5]$. The empirical speedup of 1.26×–2.13× reflects that iteration count follows a logarithmic (not linear) relationship with initial error.

The value $(1-\alpha) = 0.021$ is small enough to provide substantial warm-start benefit, yet large enough to prevent the initial guess from drifting arbitrarily far from x_{ref} when the system undergoes a structural change (topology change, regime shift). This is analogous to a biological organism that retains memory of its environment but periodically refreshes its priors.

B. Limitations

BAWS-NR assumes slow temporal variation ($\delta < D$). When the system changes abruptly (e.g., sudden large disturbances), the warm-start may provide little benefit or require a reset to flat-start. The inZORi framework handles this via an automatic fallback mechanism. Additionally, the specific value $\alpha = 0.979$ was evolved on the 1354-bus power flow problem; while it generalizes empirically, a domain-specific evolution may yield slightly better results for problems with very different temporal dynamics (e.g., financial markets).

C. Computational Overhead

The BAWS-NR warm-start computation is $x_0 = \alpha \cdot x^*(t-1) + (1-\alpha) \cdot x_{ref}$, which costs $O(n)$ scalar multiplications and additions. For power systems with $n = 1354$, this is ~2,708 floating-point operations compared to $O(n^3)$ for the LU factorization of the Jacobian. The overhead is negligible in all tested domains.

VI. CONCLUSION

We have presented BAWS-NR, a universal warm-start strategy for Newton-Raphson solvers in sequential nonlinear systems. The method is characterized by a single parameter $\alpha = 0.979$, requires no domain-specific tuning, and is guaranteed to never worsen convergence when the temporal variation is smaller than the flat-start distance. Empirical validation across 142,056 converged NR solves in six independent domains yields a consistent mean speedup of 1.59×, in close agreement with the quadratic convergence model prediction of ~2.5 iterations saved (empirical: 1.95 ± 0.87).

The primary practical impact is in real-time power systems operation: a 1.66× speedup in N-1 contingency screening translates directly to 66% more security checks per minute, enabling higher renewable energy penetration without reducing grid situational awareness. The universality demonstrated here suggests BAWS-NR may be applicable as a drop-in improvement to any Newton-Raphson solver operating on time-varying systems.

Future work will explore domain-specific α evolution for financial markets and investigate BAWS-NR in the context of nonlinear model predictive control.

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